

VueJS Server-Side Template XSS

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Content

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README.md

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Vue.js serverside template XSS example

This repository demonstrates how web apps that use both serverside rendering and Vue.js can be vulnerable to XSS even if they take precautions.

`index.php` is a vulnerable PHP script. `fix-v-pre.php` and `fix-servervars-global.php` are fixed versions of the vulnerable script. The rest of this README walks through how to exploit the vulnerability, shows how to fix the vulnerability and then discusses the scope and impact of such a vulnerability.

Note that this vulnerability is not specific to PHP nor is it specific to Vue.js. If you have an app that mixes serverside rendering with clientside rendering, you might be vulnerable.

I suggest that you run the demo, try to exploit it and then try to fix it. It's a great learning experience.

Running the demo

1. Install [docker & docker-compose](#)
2. Run the demo

```
docker-compose up -d
```

1. Open `<http://localhost:8080>` in your browser

If you don't want to bother with `docker` you can throw the `index.php` file on a PHP capable server and host it there. Keep in mind that this file is vulnerable to XSS by design, you should run this on a local environment.

Walkthrough

:rotating_light: Warning: spoilers ahead! If you want to exploit this yourself, stop reading. The following gives away the solution. :rotating_light:

First we open the app and see that we have two things to play with:

1. a textbox that lets you inject text
2. a counter application

When we play around with the textbox, see that whatever we put in there ends up as a query parameter. If we put `foobar` in the textbox, we end up with the following url: `http://localhost:8080/?injectme=foobar`. We can also see that the text `foobar` is injected in the page.

This injection looks to be done serverside. We can confirm this by looking at the source of the page. We see that `foobar` was part of the response sent by the server. We have an opportunity for XSS here. Let's try the usual bag of tricks.

When we try `<script>alert('xss')</script>` we get `<script>alert('xss')</script>`. Similarly, if we try `` we get `<img src="nope.jpg" onerror="alert('xss')"/>`.

Looks like everything gets escaped properly. We can confirm this by looking at the source code of the page. Back to the drawing board. Let's try to understand how this app works. How does the counter app work? Maybe it can help us here.

The counter app is built with Vue.js. For those who are not familiar, Vue.js is a javascript lib that runs in the browser. It lets you build dynamic frontend apps.

One way of using Vue.js is to write a template in the HTML of your page and then tell Vue.js to render it through javascript. This is a common thing to do when you have an application that uses serverside rendering and you want to add some dynamism to it. This is exactly what's happening here.

When we look at the template, we see that our injected value from before is rendered directly inside of the template.

```
<div id="injectable-app">
  <div>
    You have injected: OUR_INJECTED_VALUE_GOES_HERE
  </div>

  <button type="button" @click="dec">-</button>
  {{counter}}
  <button type="button" @click="inc">+</button>
</div>
```

Maybe we can exploit this. Vue.js templates allow you to use expressions. Expressions are bits of code that take some data, transform it and output it. They're basically javascript. In Vue.js expressions have the form `{{ ... code goes here ... }}`.

Let's give this a shot. We write `{{ 2 + 2 }}` in the textbox and we get `You have injected: 4`. It works! :tada:

For good measure we'll do an `alert` to prove that we have full control. We put `{{ alert('xss') }}` in the textbox and nothing. The whole counter is gone. When we look at the console we get an error:

```
TypeError: alert is not a function
    at Proxy.eval (eval at createFunction (vue.js:10518), <anonymous>:2:114)
    at Vue$3.Vue._render (vue.js:4465)
    at Vue$3.updateComponent (vue.js:2765)
    at Watcher.get (vue.js:3113)
    at new Watcher (vue.js:3102)
    at mountComponent (vue.js:2772)
    at Vue$3.$mount (vue.js:8416)
    at Vue$3.$mount (vue.js:10777)
    at Vue$3.Vue._init (vue.js:4557)
    at new Vue$3 (vue.js:4646)
```

`alert` is not a function? What's going on here? Vue.js expressions are evaluated in the context of the `Vue` instance that they are rendered with. In other words, when we try to render `{{ foobar }}`, it looks for the `foobar` property in the template data. When we wrote `{{ alert('xss') }}` it was interpreted as `templateData.alert('xss')`. We got the error because our template data does not have a property named `alert`.

You can think of this as being stuck inside a javascript jail or sandbox. It's important to note that Vue.js doesn't have a real sandbox. It doesn't actively try to prevent you from accessing stuff outside the template data. This is just a side effect of how it evaluates expressions.

How do we get out of this "sandbox"? There are many ways. If you want to flex your javascript muscles you can give it a shot. The solution I went with is:

```
{{ constructor.constructor("alert('xss')")() }}
```

This looks obtuse but it's surprisingly simple. We know that we're evaluated against our template data. When we write `constructor`, it's interpreted as `templateData.constructor`. Our template data is an object. All objects in javascript have a constructor. So `constructor` gives us `Vue$3` (the Vue.js constructor).

In javascript, all constructors are functions and all functions are objects. This means that `Vue$3` has a constructor. This constructor is the `Function` constructor. Writing `constructor.constructor` gives us the `Function` constructor.

The `Function` constructor lets us define a function dynamically at runtime. We pass it the code of our function and it returns a function that we can run. In this case we end up with `Function("alert('xss')")()`. This creates a function that calls `alert` (the real `alert` in the global scope) and then calls it.

That's it. We've made it, we have injected javascript in a page we don't control and this javascript has access to the global scope. At this point we can do anything the browser can. This is full blown XSS.

Why does this work?

This exploit is possible because the app is mixing server-side rendering and client-side rendering.

In this case, we have our PHP app that takes user input (a query parameter) and uses it to render an HTML page. The app escapes the input for HTML entities ensuring that simple XSS is impossible. When the page makes it into the browser, Vue.js takes part of this HTML and renders it like a template. We've seen that it's basically doing a complex `eval` on that HTML.

In this context, Vue.js can't tell the difference between the user input which may not be safe and the template which is essentially code and is considered safe.

When it's able to tell the difference between user input and template code, Vue.js does an amazing job at preventing XSS. In fact, it does a better job than PHP because it will treat its template data as dangerous by default and always escape it. You don't have to remember to escape your data.

How can I protect against this?

The simple fix for this is to use the `v-pre` directive whenever you are injecting server-side values into a client-side template.

In this case, you would change

```
<div id="injectable-app">
  <div>
    You have injected:
    <?= htmlspecialchars($_GET['injectme']) ?>
  </div>

  <button type="button" @click="dec">-</button>
  {{counter}}
  <button type="button" @click="inc">+</button>
</div>
```

to

```
<div id="injectable-app">
  <div v-pre>
    You have injected:
    <?= htmlspecialchars($_GET['injectme']) ?>
  </div>

  <button type="button" @click="dec">-</button>
  {{counter}}
  <button type="button" @click="inc">+</button>
</div>
```

While this solution does work, it's not great. It's easy for anyone to forget to use the `v-pre` directive. If a single developer forgets to do this, you're screwed all over again.

When it comes to security, I prefer systematic solutions. A better solution would be to define a global variable in the page that holds all serverside variables. This does not prevent a developer from mixing serverside and clientside rendering but it does give them a secure mechanism for passing values from the server to the client.

We can implement this like so:

```

<div id="injectable-app">
  <div>
    You have injected: {{ SERVER_VARS.injectMe }}
  </div>

  <button type="button" @click="dec">-</button>
  {{counter}}
  <button type="button" @click="inc">+</button>
</div>
<script src="https://cdn.jsdelivr.net/npm/vue@2.5.13/dist/vue.js"></script>
<?php
$serverVars = [
  'injectMe' => (string) $_GET['injectme']
];
?>
<script>
window.SERVER_VARS = JSON.parse(atob('<?=' . base64_encode(json_encode($serverVars)) . '?>'));
Vue.prototype.SERVER_VARS = window.SERVER_VARS;
</script>

```

The full fix is available in `fix-servervars-global.php`.

Is this a real threat?

After reading this, you might wonder: Why would anyone in their right minds mix sever-side and client-side rendering?

I think that it's pretty reasonable for a developer to add Vue.js to their existing server-side rendering app and think that everything is going to be fine. Vue.js advertises itself as a "progressive frameworks". They expect you to do this. Also, the security risks are not immediately apparent. Getting our XSS to run was pretty roundabout.

If you do a little googling, you'll find a bunch of examples and tutorials on how to use Vue.js with other server-side rendering frameworks. While I don't have the numbers to back this, I think that there are plenty of apps out there that mix server-side rendering and client-side rendering. All of those apps could be vulnerable to this kind of XSS.

Is this specific to PHP?

Not at all. This can work with any server-side language or technology. You can take this example and rewrite it in any server-side technology and it would still be vulnerable. It doesn't matter how much automatic escaping this technology does, because no one automatically escapes variables injected in Vue.js templates.

What about other frameworks / libs?

Any library or framework that lets you write templates in HTML is potentially vulnerable to this.

Angular 1 apps are famously vulnerable to this. In their [security guide](#) the angular team warns explicitly about this.

Generating AngularJS templates on the server containing user-provided content. This is the most common pitfall where you are generating HTML via some server-side engine such as PHP, Java or ASP.NET.

They've also tried for a long time to build a sandbox that mitigates against XSS coming from the server. Every time that they've improved or fixed the sandbox, it was broken or bypassed. Eventually the angular team [got rid of the sandbox](#).

Frameworks like React and Angular 2+ are not likely to be vulnerable to this kind of attack because they don't let you write templates in HTML and they force the use of a compiler. This makes injecting user input from the server into clientside templates very unlikely. I'm not saying that React and Angular 2+ are bullet proof. You just have to try harder to be vulnerable.

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